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WHERE TO NEXT?

Finding the ideal spot(s) for possible NFL expansion

1. INTRODUCTION

The National Football League is undisputably the most popular league in America, with reportedly over 400 million fans across the states. As the NFL continues to grow, it's only a matter of time before we see the league expand, which would mean more teams joining the league. Today we are going to look at the best possible relocation cities should the possibility of expansion becomes a reality.
The steps we will take go as follows:

- Current NFL city data
- Model
- Interpretation
- Possible Expansion City Data
- Where to next?

2. NFL CITY DATA

We are aiming to measure the effectiveness on how non-football factors (aside from win%) effect "Success". Success variable will be defined as a mix of Value, Attendance percentage, and Revenue variables. The independent variables go as follows:

- Market: Local TV reach and viewership
- Population: Proper NOT METRO population
- Competition: How many other pro teams (specifically NBA, MLB, NHL) teams are in the city

- Proximity: How close the city is to the nearest NFL stadium
- Win%: Team all time win percentage

We are going to input our variables into an Excel table, sources for our values will be at the end of the paper.

Team	Proximity	Competition	Win%	Population (in millions)	Market (in millions)	Value (in billions)	Attendance	Revenue (in billions)
49ers	310	4	0.547	4.7	2.653	8.6	103.90%	0.622
Bears	150	5	0.554	9.5	3.472	8.2	94.50%	0.556
Bengals	100	1	0.455	2.2	0.926	5.25	99.10%	0.498
Bills	170	1	0.494	1.2	0.613	5.95	99%	0.503
Broncos	550	3	0.528	3	1.798	6.8	100.40%	0.563
Browns	90	2	0.503	2.2	1.512	6.4	95.30%	0.545
Buccaneers	180	2	0.412	3.2	2.035	6.6	96.90%	0.531
Cardinals	230	2	0.419	4.8	2.158	5.5	100.20%	0.5
Chargers	230	6	0.502	13.2	5.735	6	102.20%	0.518
Chiefs	410	1	0.55	2.2	0.986	6.2	96.10%	0.54
Colts	100	1	0.524	2.1	1.183	5.9	103.70%	0.545
Commanders	30	4	0.495	6.3	2.556	7.6	95%	0.545
Cowboys	230	3	0.571	7.6	2.963	13	116.20%	1.14
Dolphins	190	3	0.55	6.1	1.693	7.5	100%	0.6
Eagles	90	3	0.502	6.2	2.997	8.3	100%	0.598
Falcons	210	2	0.438	6.1	2.649	6.35	99.40%	0.544
Giants	90	7	0.518	20.1	7.453	10.1	96.70%	0.639
Jaguars	180	0	0.429	1.6	0.757	5.6	91.00%	0.517
Jets	90	7	0.432	20.1	7.453	8.1	90.80%	0.56
Lions	90	3	0.462	4.4	1.863	5.4	99.40%	0.495
Packers	180	0	0.572	0.3	0.456	6.65	95.60%	0.577
Panthers	230	1	0.454	2.7	1.291	5.7	96.70%	0.53

Patriots	170	3	0.555	4.9	2.49	9	100%	0.684
Raiders	230	2	0.509	2.3	0.834	7.7	95.80%	0.729
Rams	230	6	0.51	13.2	5.735	10.5	104.60%	0.686
Ravens	30	4	0.57	2.8	1.13	6.1	99%	0.544
Saints	300	1	0.463	1	0.664	5.3	94.40%	0.533
Seahawks	720	2	0.525	4	2.099	6.7	100%	0.555
Steelers	110	2	0.538	2.5	1.166	6.5	96.90%	0.548
Texans	230	2	0.449	7.1	2.57	7.4	98.10%	0.605
Titans	210	1	0.477	2	1.102	6.3	85.20%	0.516
Vikings	270	3	0.55	3.7	1.887	6.25	100.30%	0.54

3. MODEL

We will be using STATA software to build our model. These are the steps in simple terms:

- Import table from Excel
- Convert Value, Attendance Percentage, and Revenue into log variables to reduce outlier influence
- Convert the log values into Z Scores to put all the variables on the same scale
- Run PCA test to create dependent variable
- Convert independent variables into logs
- Run regression test

```
. reg success_pc1 l_prox l_comp l_win l_pop l_market

      Source |       SS           df       MS      Number of obs   =        30
-------------+----------------------------------   F(5, 24)        =      4.04
       Model |  30.3776572         5  6.07553145   Prob > F        =    0.0084
    Residual |  36.0850069        24  1.50354196   R-squared       =    0.4571
-------------+----------------------------------   Adj R-squared   =    0.3440
       Total |  66.4626642        29  2.29181601   Root MSE        =    1.2262

------------------------------------------------------------------------------
 success_pc1 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      l_prox |   .2537386   .3558506     0.71   0.483     -.480701    .9881781
      l_comp |  -.6220031   .8643416    -0.72   0.479    -2.405916     1.16191
       l_win |   8.445617     2.8885     2.92   0.007     2.484045    14.40719
       l_pop |    1.08671   1.227771     0.89   0.385    -1.447285    3.620704
    l_market |   .3434501   1.486341     0.23   0.819    -2.724206    3.411106
       _cons |   3.372589   3.446726     0.98   0.338    -3.741104    10.48628
```

4. INTERPRETATION

The 2 main results we care about for this particular model are Prob > F and R-squared. If Prob > F is greater than .05, our model would be statistically insignificant, which would NOT be good. Luckily for us, our number is .0084, which means our model IS statistically significant. R-Squared measures how well our independent variables explain dependent variable variation. Our rounded number is .46, which is a little bit below average. This might sound bad, and while it could certainly be better, some context is needed. Essentially what we are trying to do is explain team success using non-football variables, which logically are not going to impact success as much as football related variables, such as number of Pro Bowlers, money spent on roster, QB Passer Rating, the list goes on. So while .46 might not sound the best, this is actually pretty solid given the context.

5. POSSIBLE EXPANSION CITY DATA

Now, it's time to see where the NFL should consider expanding. We are going to be looking at 10 cities across the U.S. with competitive TV markets, population, competition, and proximity compared to current NFL cities. For Win% variable, we are going to set it to .5, which is league average.

City	Proximity	Competition	Win%	Population (in millions)	Market (in millions)
Austin	160	0	0.5	1	1
San Antonio	200	1	0.5	1.5	1.1
St. Louis	240	2	0.5	0.29	1.3
Orlando	140	1	0.5	0.32	1.9
Salt Lake City	420	2	0.5	0.2	1.1
Oakland	40	0	0.5	0.44	2.6
San Diego	120	1	0.5	1.4	1.1
Portland	170	1	0.5	0.64	1.3
Columbus	110	0	0.5	0.91	1

Oklahoma City	210	1	0.5	0.7	0.76

6. WHERE TO NEXT?

Like earlier, we are going to import our Excel table into STATA, and convert our independent variables into log variables. Then we are going to predict Success variables based on our independent variables.

```
. gsort -success_hat5

. list Team success_hat5

     +-----------------------------+
     |           Team   success~5  |
     |-----------------------------|
  1. | Salt Lake City    1.198979  |
  2. |      St. Louis    .8961732  |
  3. |        Orlando    .7366729  |
  4. |    San Antonio    .7191899  |
  5. |       Portland    .6667817  |
     |-----------------------------|
  6. |  Oklahoma City    .4800704  |
  7. |         Austin    .4676537  |
  8. |      San Diego    .3246043  |
  9. |       Columbus    .1762948  |
 10. |        Oakland   -.0034592  |
     +-----------------------------+
```

According to our prediction Salt Lake City would be the best destination for a NFL expansion team, followed by St. Louis, Orlando, and San Antonio. Salt Lake City has proven to have strong fan bases among their NBA and NHL teams, aswell as one of the most passionate best college football fanbase (BYU). St. Louis previously had the Rams before they reloacted to LA, so it's proven the city can have a successful franchise. Orlando and San Antonio both have NBA teams with great fanbases, which could translate to pro football. The NFL will add more teams, and only time will tell where said teams will find their homes.

7. REFERNCES

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